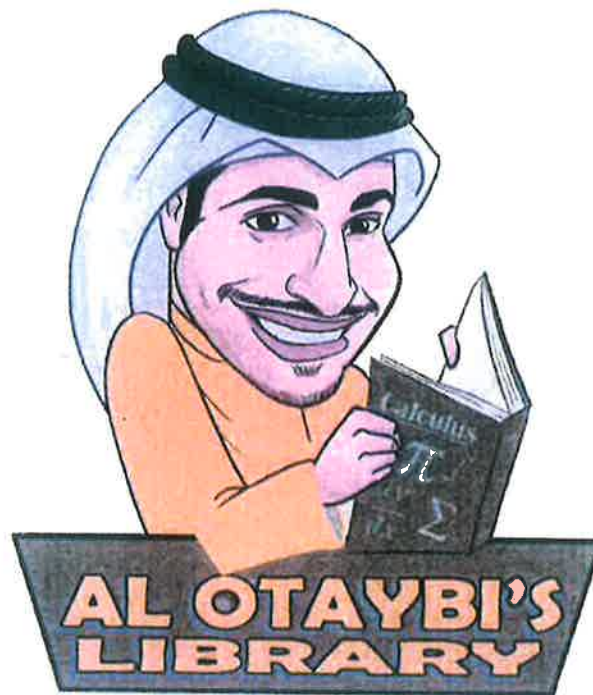


Kuwait University

2021/2022

Spring course



Calculus B

First

Section : 7.2

7.2 Trigonometric Integrals.

(4)

Eng. Mohammad Alotaybi 90048657 Twitter: @M_Alotaybi



Calculus B Quiz # 3



$$\int \csc x \, dx$$

$$\int e^x \tan(e^x) \, dx$$

$$\int \frac{\csc x \cot x}{5 \csc x + 7} \, dx$$

$$\int x \cos x \, dx$$



**AL OTAYBI'S
LIBRARY**



أنواع التكاملات



By parts

عند وجود أولوية من
الأولويات



u

dv



بس دوال مثلثية
أو (h)



مراجعة على طريقة التعامل مع الزوايا



$$\sin(-x) = -\sin x$$

$$\tan(-x) = -\tan x$$

$$\cos(-x) = \cos x$$

cos

تبلغ السالب بزوايتها



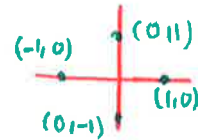
Find

$$\sin(5\pi) = \sin(\pi) = 0$$

$$\rightarrow 5\pi - 2\pi = 3\pi - 2\pi = \pi$$

$$\cos(3\pi) = \cos(\pi) = -1$$

$$\rightarrow 3\pi - 2\pi = \pi$$



Find

$$\int_0^{\frac{\pi}{2}} \sin\left(-\frac{x}{2}\right) dx$$

$$= -\int_0^{\frac{\pi}{2}} \sin\left(\frac{x}{2}\right) dx$$

$$+ \left[\frac{\cos\left(\frac{x}{2}\right)}{-\frac{1}{2}} \right]_0^{\frac{\pi}{2}} = 2 \left[\cos\left(\frac{\pi}{4}\right) - \cos(0) \right]$$

$$= 2 \left[\frac{1}{\sqrt{2}} - 1 \right]$$

تكاليف الدوال المثلثية



$$\sin 2x = 2 \sin x \cos x$$

$$\cos^2 x - \sin^2 x$$

$$\cos 2x = \frac{\cos^2 x - \sin^2 x}{2 \cos^2 x - 1}$$

$$1 - 2 \sin^2 x$$

$$\cos^2 x = \frac{1}{2} (1 + \cos 2x)$$

$$\sin^2 x = \frac{1}{2} (1 - \cos 2x)$$

AL OTAYBI'S LIBRARY



رفيقات أو بروجهم

$\sin x, \cos x$

$\sec x, \tan x$

$\tan x$

$\csc x, \cot x$

$\cot x$

$$\sin(a) \sin(b) = \frac{1}{2} [\cos(a-b) - \cos(a+b)]$$

$$\cos(a) \cos(b) = \frac{1}{2} [\cos(a-b) + \cos(a+b)]$$

$$\sin(a) \cos(b) = \frac{1}{2} [\sin(a-b) + \sin(a+b)]$$



رفيجات أو بروحهم

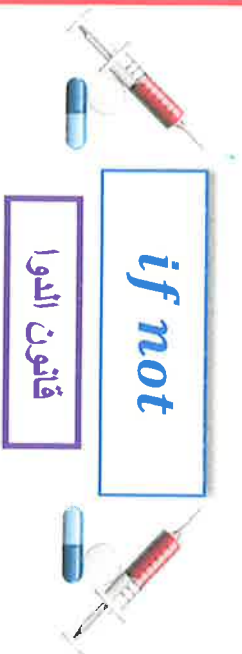


$\sin x, \cos x$

(1) أاور الفردي , وأطلع منه وحدة.

(2) أحوله.

(3) أفرض u الدالة الثانية.



if not

قانون الدواء

$$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$$

$\sec x, \tan x$
 $\csc x, \cot x$

$\tan x$
 $\cot x$

$\sec x, \tan x$
 $\csc x, \cot x$

زوجي

فردي

أبي

AL OTAYBI'S LIBRARY

(1) أطلع شتين
(2) أحول الي طلعتة.
(3) أضرب.



قوانين التحويل



$$\cos^2 x + \sin^2 x = 1$$

$$1 + \tan^2 x = \sec^2 x$$

$$\cot^2 x + 1 = \csc^2 x$$

$$\cosh^2 x - \sinh^2 x = 1$$

$$1 - \tanh^2 x = \operatorname{sech}^2 x$$

$$\coth^2 x - 1 = \operatorname{csch}^2 x$$

$$\int \sinh^3 x \cosh^2 x \, dx$$

$$\int \sinh^2 x \cosh^2 x \sinh x \, dx$$

$$\int (\cosh^2 x - 1) \cosh^2 x \sinh x \, dx$$

Let $u = \cosh x$
 $du = \sinh x \, dx$

$$\int (u^2 - 1) u^2 \, du$$

$$\int (u^4 - u^2) \, du$$

$$= \frac{u^5}{5} - \frac{u^3}{3} + C$$

$$= \frac{(\cosh x)^5}{5} - \frac{(\cosh x)^3}{3} + C$$

يتحول !!

$u \leftarrow$ الثانية





Text Book Question



$$\int \sqrt{\cos \theta} \sin^3 \theta \, d\theta$$

$$\int (\cos \theta)^{\frac{1}{2}} \sin^2 \theta \sin \theta \, d\theta$$

$$\int (\cos \theta)^{\frac{1}{2}} (1 - \cos^2 \theta) \sin \theta \, d\theta$$

$$\text{Let } u = \cos \theta$$

$$du = -\sin \theta \, d\theta$$

$$-du = \sin \theta \, d\theta$$

$$- \int u^{\frac{1}{2}} (1 - u^2) \, du$$

$$- \int (u^{\frac{1}{2}} - u^{\frac{5}{2}}) \, du$$

$$= - \left[\frac{2}{3} u^{\frac{3}{2}} - \frac{2}{7} u^{\frac{7}{2}} \right] + C$$

$$= - \left[\frac{2}{3} (\cos \theta)^{\frac{3}{2}} - \frac{2}{7} (\cos \theta)^{\frac{7}{2}} \right] + C$$

$$\int \sin^5 x \cos^3 x \, dx$$

Nov 2021

$$\int \sin^5 x \cos^2 x \cos x \, dx$$

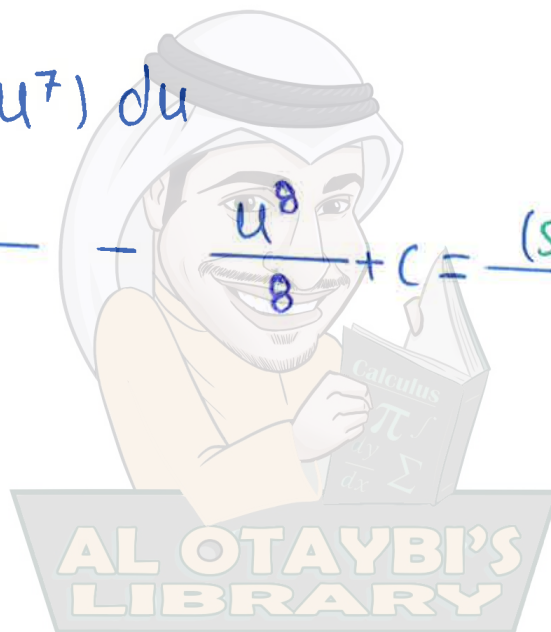
$$\int \sin^5 x (1 - \sin^2 x) \cos x \, dx$$

$$\text{Let } u = \sin x \\ du = \cos x \, dx$$

$$\int u^5 (1 - u^2) \, du$$

$$\int (u^5 - u^7) \, du$$

$$= \frac{u^6}{6} - \frac{u^8}{8} + C = \frac{(\sin x)^6}{6} - \frac{(\sin x)^8}{8} + C$$



$$\int \sinh^2(6x) \cosh^5(6x) dx$$

$$\int \sinh^2(6x) \cosh^4(6x) \cosh(6x) dx$$

$$(\cosh^2(6x))^2$$

$$\int \sinh^2(6x) (1 + \sinh^2(6x))^2 \cosh(6x) dx$$

Let $u = \sinh(6x)$
 $du = 6 \cosh(6x) dx$

$$\frac{1}{6} du = \cosh(6x) dx$$

$$\frac{1}{6} \int u^2 (1 + u^2)^2 du$$

$$\frac{1}{6} \int u^2 (1 + 2u^2 + u^4) du$$

$$\frac{1}{6} \int (u^2 + 2u^4 + u^6) du$$

$$\frac{1}{6} \left[\frac{u^3}{3} + 2 \frac{u^5}{5} + \frac{u^7}{7} \right] + C$$

$$\frac{1}{6} \left[\frac{(\sinh 6x)^3}{3} + \frac{2}{5} (\sinh 6x)^5 + \frac{1}{7} (\sinh 6x)^7 \right] + C$$

$$\int \frac{\sin^5 x}{\sec^3 x} dx$$

لازم رفيجات !

$$\int \sin^{\frac{5}{2}} x \cos^3 x dx$$

صاروا رفيجات



$$\int \sin^{\frac{5}{2}} x \cos^2 x \cos x dx$$

الحين
طبق الخطوات

$$\int \sin^{\frac{5}{2}} x (1 - \sin^2 x) \cos x dx$$

Let $u = \sin x$
 $du = \cos x dx$

$$\int u^{\frac{5}{2}} (1 - u^2) du$$

$$\int (u^{\frac{5}{2}} - u^{\frac{9}{2}}) du$$

$$= \frac{u^{\frac{7}{2}}}{\frac{7}{2}} - \frac{u^{\frac{11}{2}}}{\frac{11}{2}} + C$$

$$= \frac{(\sin x)^{\frac{7}{2}}}{\frac{7}{2}} - \frac{(\sin x)^{\frac{11}{2}}}{\frac{11}{2}} + C$$

$$\int \sin^3 x \sqrt[3]{\sec x} dx$$

لازم رفيجات !

$$\int \sin^3 x \frac{1}{\sqrt[3]{\cos x}} dx$$

صاروا رفيجات

$$\int \sin^3 x (\cos x)^{-\frac{1}{3}} dx$$

الحين
طبق الخطوات



$$\int \sin^2 x (\cos x)^{-\frac{1}{3}} \cdot \sin x dx$$

$$\int (1 - \cos^2 x) (\cos x)^{-\frac{1}{3}} \sin x dx$$

Let $u = \cos x$
 $du = -\sin x dx$
 $-du = \sin x dx$

$$- \int (1 - u^2) u^{-\frac{1}{3}} du$$

$$- \int (u^{-\frac{1}{3}} - u^{\frac{5}{3}}) du$$

$$- \left[\frac{u^{\frac{2}{3}}}{\frac{2}{3}} - \frac{u^{\frac{8}{3}}}{\frac{8}{3}} \right] + C$$

$$- \left[\frac{3}{2} (\cos x)^{\frac{2}{3}} - \frac{3}{8} (\cos x)^{\frac{8}{3}} \right] + C$$

$$\int \tan^2 x \cos^5 x \, dx$$

لازم
رفیقات!

$$\int \frac{\sin^2 x}{\cos^2 x} \cos^5 x \, dx$$

$$\int \sin^2 x \cos^3 x \, dx$$

$$\int \sin^2 x \cos^2 x \cos x \, dx$$

$$\int \sin^2 x (1 - \sin^2 x) \cos x \, dx$$

Let $u = \sin x$
 $du = \cos x \, dx$



$$\int u^2 (1 - u^2) \, du$$

$$\int (u^2 - u^4) \, du$$

$$= \frac{u^3}{3} - \frac{u^5}{5} + C$$

$$= \frac{(\sin x)^3}{3} - \frac{(\sin x)^5}{5} + C$$

$$\int \frac{\sin^2 x \cos^5 x}{\cot x} dx$$

Fall 2021

$$\int \frac{\sin^2 x \cancel{\cos^5 x}}{\cancel{\cos x} \sin x} dx$$

$$\int \sin^3 x \cos^4 x dx$$

$$\int \sin^2 x \cos^4 x \sin x dx$$

$$\int (1 - \cos^2 x) \cos^4 x \sin x dx$$

Let $u = \cos x$
 $du = -\sin x dx$
 $-du = \sin x dx$

$$- \int (1 - u^2) u^4 du$$

$$- \int (u^4 - u^6) du$$

$$- \left[\frac{u^5}{5} - \frac{u^7}{7} \right] + C$$

$$- \left[\frac{(\cos x)^5}{5} - \frac{(\cos x)^7}{7} \right] + C$$



Text Book Questions



$$\int \cot x \cos^2 x \, dx$$

$$\int \frac{\cos x}{\sin x} \cos^2 x \, dx$$

$$\int \cos^3 x (\sin x)^{-1} \, dx$$

$$\int \cos^2 x (\sin x)^{-1} \cos x \, dx$$

$$\int (1 - \sin^2 x) (\sin x)^{-1} \cos x \, dx$$

$$\text{Let } u = \sin x \\ du = \cos x \, dx$$

$$\int (1 - u^2) u^{-1} du = \int (u^{-1} - u) du = \ln|u| - \frac{u^2}{2} + C \\ = \ln|\sin x| - \frac{\sin^2 x}{2} + C$$

$$\int \tan^2 x \cos^3 x \, dx$$

$$\int \frac{\sin^2 x}{\cos^2 x} \cos^3 x \, dx$$

$$\int \sin^2 x \cos x \, dx$$

$$\text{Let } u = \sin x \\ du = \cos x \, dx$$

$$\int u^2 \, du$$

$$= \frac{u^3}{3} + C$$

$$= \frac{(\sin x)^3}{3} + C$$

$$\int \cos x \sin^4(2x) dx$$

$$\int \cos x (2 \sin x \cos x)^4 dx$$

$$\int \cos x 2^4 \sin^4 x \cos^4 x dx$$

$$16 \int \sin^4 x \cos^5 x dx$$

$$16 \int \sin^4 x \cos^4 x \cos x dx$$

$$16 \int \sin^4 x (\cos^2 x)^2 \cos x dx$$

Let $u = \sin x$
 $du = \cos x dx$

$$16 \int u^4 (1-u^2)^2 du$$

$$16 \int u^4 (1-2u^2+u^4) du$$

$$16 \int (u^4 - 2u^6 + u^8) du$$

$$16 \left[\frac{u^5}{5} - 2 \frac{u^7}{7} + \frac{u^9}{9} \right] + C$$

$$16 \left[\frac{(\sin x)^5}{5} - 2 \frac{(\sin x)^7}{7} + \frac{(\sin x)^9}{9} \right] + C$$

$$\int \csc^3(2x) \sin^6 x \, dx$$

$$\int \frac{1}{\sin^3(2x)} \cdot \sin^6 x \, dx$$

$$\int \frac{1}{(2 \sin x \cos x)^3} \sin^6 x \, dx$$

$$\int \frac{1}{8 \sin^3 x \cos^3 x} \sin^6 x \, dx$$

$$\frac{1}{8} \int \frac{\sin^3 x}{\cos^3 x} \, dx$$

$$\frac{1}{8} \int \sin^3 x (\cos x)^{-3} \, dx$$

$$\frac{1}{8} \int \sin^2 x (\cos x)^{-3} \sin x \, dx$$

$$\frac{1}{8} \int (1 - \cos^2 x) (\cos x)^{-3} \sin x \, dx$$

Let $u = \cos x$

$du = -\sin x \, dx \longrightarrow -du = \sin x \, dx$

$$-\frac{1}{8} \int (1 - u^2) u^{-3} \, du$$

$$-\frac{1}{8} \int (u^{-3} - u^{-1}) \, du = -\frac{1}{8} \left(\frac{u^{-2}}{-2} - \ln|u| \right) + C$$

$$= -\frac{1}{8} \left(\frac{(\cos x)^{-2}}{-2} - \ln|\cos x| \right) + C$$

$$\int \sin^3 x \, dx$$

عادي بروحها



$$\int \sin^2 x \sin x \, dx$$

$$\int (1 - \cos^2 x) \sin x \, dx$$

Let $u = \cos x$

$$du = -\sin x \, dx$$

$$-du = \sin x \, dx$$

$$-\int (1 - u^2) \, du$$

$$-\left[u - \frac{u^3}{3}\right] + C \Rightarrow -\left[\cos x - \frac{(\cos x)^3}{3}\right] + C$$

$$\int \cos^5 x \, dx$$

$$\int \cos^4 x \cos x \, dx$$

$$\int (1 - \sin^2 x)^2 \cos x \, dx$$

Let $u = \sin x$

$$du = \cos x \, dx$$

$$\int (1 - u^2)^2 \, du$$

$$\int (1 - 2u^2 + u^4) \, du$$

$$u - \frac{2u^3}{3} + \frac{u^5}{5}$$

+C

$$\sin x - \frac{2}{3}(\sin x)^3$$

$$+ \frac{1}{5}(\sin x)^5$$

+C



Nice Questions !!



$$\int \cos^5 x \sin^3 x \, dx$$

إذا اتيناهم
قريباً



يفضل
نشتغل
على
الأسفل

$$\int \cos^5 x \sin^2 x \sin x \, dx$$

$$\int \cos^5 x (1 - \cos^2 x) \sin x \, dx$$

Let $u = \cos x$
 $du = -\sin x \, dx$
 $-du = \sin x \, dx$

$$-\int u^5 (1 - u^2) \, du$$

$$-\int (u^5 - u^7) \, du = -\left[\frac{u^6}{6} - \frac{u^7}{7}\right] + C$$

$\swarrow \searrow$
 $\cos x$

$$\int \sin x \cos^{10} x \, dx$$

إذا جددنا
ان القريب
أسفل
"1"

Let $u = \cos x$
 $du = -\sin x \, dx$
 $-du = \sin x \, dx$

$$-\int u^{10} \, du$$

$$-\frac{u^{11}}{11} + C$$

$$-\frac{(\cos x)^{11}}{11} + C$$

على
الأسفل
"4"



Text Book Question



$$\int_0^{\frac{\pi}{2}} \sin^7 \theta \cos^5 \theta \, d\theta$$

$$\int_0^{\frac{\pi}{2}} \sin^6 \theta \cos^4 \theta \cos \theta \, d\theta$$

$(\cos^2 \theta)^2$

$$\int_0^{\frac{\pi}{2}} \sin^6 \theta (1 - \sin^2 \theta)^2 \cos \theta \, d\theta$$

Let $u = \sin \theta$
 $du = \cos \theta \, d\theta$

$$\int u^6 (1 - u^2)^2 du = \int u^6 (1 - 2u^2 + u^4) du$$

$$= \int (u^6 - 2u^8 + u^{10}) du$$

$$= \frac{u^7}{7} - 2 \frac{u^9}{9} + \frac{u^{11}}{11}$$

$$= \left[\frac{(\sin \theta)^7}{7} - \frac{2(\sin \theta)^9}{9} + \frac{(\sin \theta)^{11}}{11} \right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{7} - \frac{2}{9} + \frac{1}{11} - 0$$

$$\int \sin^2 x \, dx$$

$$\frac{1}{2} \int (1 - \cos 2x) \, dx$$

$$\frac{1}{2} \left[x - \frac{\sin 2x}{2} \right] + C$$



$$\int \sin^4 x \, dx$$

$$\int (\sin^2 x)^2 \, dx$$

$$\int \left(\frac{1}{2} (1 - \cos 2x) \right)^2 \, dx$$

$$\int \frac{1}{4} (1 - \cos 2x)^2 \, dx$$

$$\frac{1}{4} \int (1 - 2\cos 2x + \cos^2 2x) \, dx$$

$$x - \frac{2 \sin 2x}{2} + C_1$$

$$\int \cos^2 2x \, dx$$

$$\frac{1}{2} \int (1 + \cos 4x) \, dx$$

$$\frac{1}{2} \left(x + \frac{\sin 4x}{4} \right) + C_2$$

$$\Rightarrow \frac{1}{4} \left[x - \frac{2 \sin 2x}{2} + \frac{1}{2} \left(x + \frac{\sin 4x}{4} \right) \right] + C$$



Text Book Question



$$\int \sin^2 x \cos^2 x \, dx$$

$$\int \frac{1}{2}(1 - \cos 2x) \frac{1}{2}(1 + \cos 2x) \, dx$$

$$\frac{1}{4} \int (1 - \cos 2x)(1 + \cos 2x) \, dx$$

$$\frac{1}{4} \int (1 - \cos^2 2x) \, dx$$

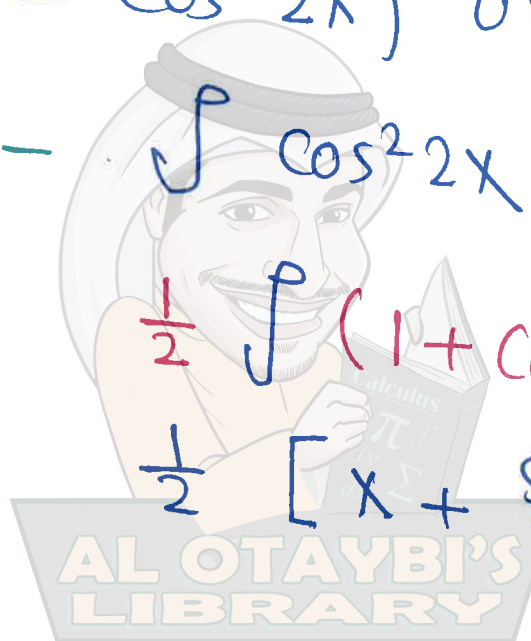
$x + C_1$

$$- \int \cos^2 2x \, dx$$

$$\frac{1}{2} \int (1 + \cos 4x) \, dx$$

$$\frac{1}{2} \left[x + \frac{\sin 4x}{4} \right] + C_2$$

$$\Rightarrow \frac{1}{4} \left[x - \frac{1}{2} \left[x + \frac{\sin 4x}{4} \right] \right] + C$$



$$\int \underline{x} \sin^2(x^2) \underline{dx}$$

Let $t = x^2$

$$dt = 2x dx \longrightarrow \frac{1}{2} dt = x dx$$

$$\frac{1}{2} \int \sin^2 t \, dt$$

$$\frac{1}{2} \cdot \frac{1}{2} \int (1 - \cos 2t) \, dt$$

$$\frac{1}{4} \left[t - \frac{\sin 2t}{2} \right] + C$$

$$\frac{1}{4} \left[x^2 - \frac{\sin 2x^2}{2} \right] + C$$

$$\int \underline{x^2} \sin^3(x^3 + 1) \cos^2(x^3 + 1) \underline{dx}$$

Let $t = x^3$

$$dt = 3x^2 dx \longrightarrow \frac{1}{3} dt = x^2 dx$$

$$\frac{1}{3} \int \sin^3(t+1) \cos^2(t+1) \, dx$$

$$\frac{1}{3} \int \sin^2(t+1) \cos^2(t+1) \sin(t+1) \, dt$$

$$\frac{1}{3} \int (1 - \cos^2(t+1)) \cos^2(t+1) \sin(t+1) \, dt$$

Let $u = \cos(t+1)$

$$du = -\sin(t+1) \, dt \longrightarrow -du = \sin(t+1) \, dt$$

$$-\frac{1}{3} \int (1 - u^2) u^2 \, du = -\frac{1}{3} \int (u^2 - u^4) \, du$$

$$-\frac{1}{3} \left(\frac{u^3}{3} - \frac{u^5}{5} \right) + C$$

$$-\frac{1}{3} \left(\frac{\cos^3(t+1)}{3} - \frac{\cos^5(t+1)}{5} \right) + C$$

$$-\frac{1}{3} \left(\frac{\cos^3(x^3+1)}{3} - \frac{\cos^5(x^3+1)}{5} \right) + C$$

$$\int \frac{\tan^3 x}{\sqrt{\sec x}} dx$$

$$\int \underline{\tan^3 x} (\sec x)^{-\frac{1}{2}} dx$$

← فزي
→ ~~لزي~~

$$\int \tan^2 x (\sec x)^{-\frac{1}{2}-1} \sec x \tan x dx$$

$$\int (\sec^2 x - 1) (\sec x)^{-\frac{3}{2}} \sec x \tan x dx$$

Let $u = \sec x$
 $du = \sec x \tan x dx$

$$\int (u^2 - 1) u^{-\frac{3}{2}} du$$

$$\int (u^{\frac{1}{2}} - u^{-\frac{3}{2}}) du$$

$$= \frac{u^{\frac{3}{2}}}{\frac{3}{2}} - \frac{u^{-\frac{1}{2}}}{-\frac{1}{2}} + C$$

$$= \frac{2}{3} (\sec x)^{\frac{3}{2}} + 2 (\sec x)^{-\frac{1}{2}} + C$$



Text Book Question



$$\int \tan^3 x \sec x \, dx$$

فردي ← $\tan^3 x$ $\rightarrow$ زوجي $\sec x$

$$\int \tan^2 x \sec x \tan x \, dx$$

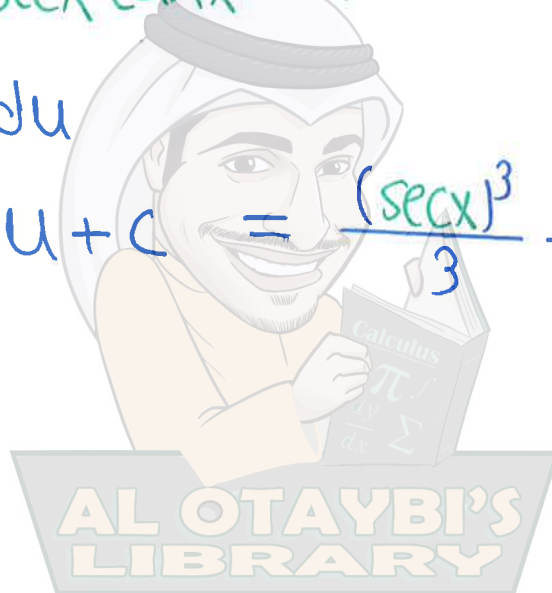
$$\int (\sec^2 x - 1) \sec x \tan x \, dx$$

$$\text{Let } u = \sec x$$

$$du = \sec x \tan x \, dx$$

$$\int (u^2 - 1) \, du$$

$$\frac{u^3}{3} - u + C = \frac{(\sec x)^3}{3} - \sec x + C$$



$$\int \tanh^5 x \operatorname{sech}^3 x \, dx$$

فرد $\xrightarrow{\quad}$ زوج $\times$

$$\int \tanh^4 x \operatorname{sech}^2 x \operatorname{sech} x \tanh x \, dx$$

$$(\tanh^2 x)^2$$

$$\int (1 - \operatorname{sech}^2 x)^2 \operatorname{sech}^2 x \operatorname{sech} x \tanh x \, dx$$

let $u = \operatorname{sech} x$
 $du = -\operatorname{sech} x \tanh x \, dx$
 $-du = \operatorname{sech} x \tanh x \, dx$

$$\int (1 - u^2)^2 u^2 \, du$$

$$\int (1 - 2u^2 + u^4) u^2 \, du$$

$$\int (u^2 - 2u^4 + u^6) \, du$$

$$= \left[\frac{u^3}{3} - 2 \frac{u^5}{5} + \frac{u^7}{7} \right] + C$$

$$= \left[\frac{(\operatorname{sech} x)^3}{3} - \frac{2}{5} (\operatorname{sech} x)^5 + \frac{(\operatorname{sech} x)^7}{7} \right] + C$$

$$\int \frac{\coth^3 x}{\sinh^3 x} dx$$

$$\int \underbrace{\coth^3 x}_{\text{فوري}} \underbrace{\csch^3 x}_{\text{زوي}} dx$$

$$\int \coth^2 x \csch^2 x \csch x \coth x dx$$

$$\int (\csch^2 x + 1) \csch^2 x \csch x \coth x dx$$

Let $u = \csch x$
 $du = -\csch x \coth x dx$
 $-du = \csch x \coth x dx$

$$-\int (u^2 + 1) u^2 du$$

$$-\int (u^4 + u^2) du$$

$$-\left[\frac{u^5}{5} + \frac{u^3}{3} \right] + C$$

$$-\left[\frac{(\csch x)^5}{5} + \frac{(\csch x)^3}{3} \right] + C$$

$$\int \frac{\sec^4 x}{\sqrt{\tan x}} dx$$

$$\int \sec^4 x (\tan x)^{-\frac{1}{2}} dx$$

زیر فری
✓ x

$$\int \sec^2 x (\tan x)^{-\frac{1}{2}} \sec^2 x dx$$

$$\int (1 + \tan^2 x) (\tan x)^{-\frac{1}{2}} \sec^2 x dx$$

Let $u = \tan x$
 $du = \sec^2 x dx$

$$\int (1 + u^2) u^{-\frac{1}{2}} du$$

$$\int (u^{-\frac{1}{2}} + u^{\frac{3}{2}}) du$$

$$= \frac{u^{\frac{1}{2}}}{\frac{1}{2}} + \frac{u^{\frac{5}{2}}}{\frac{5}{2}} + C$$

$$= 2 u^{\frac{1}{2}} + \frac{2}{5} u^{\frac{5}{2}} + C$$

$$= 2 (\tan x)^{\frac{1}{2}} + \frac{2}{5} (\tan x)^{\frac{5}{2}} + C$$

$$\int \frac{\csc^4(\pi x)}{\sqrt[3]{\cot(\pi x)}} dx$$

نقطة $\int \csc^4(\pi x) (\cot(\pi x))^{-\frac{1}{3}} dx$

$$\int \csc^2(\pi x) (\cot(\pi x))^{-\frac{1}{3}} \csc^2(\pi x) dx$$

$$\int (\cot^2(\pi x) + 1) (\cot(\pi x))^{-\frac{1}{3}} \csc^2(\pi x) dx$$

Let $u = \cot(\pi x)$
 $du = -\pi \csc^2(\pi x) dx$
 $\div (-\pi)$

$$-\frac{1}{\pi} du = \csc^2(\pi x) dx$$

$$-\frac{1}{\pi} \int (u^2 + 1) u^{-\frac{1}{3}} du$$

$$-\frac{1}{\pi} \int (u^{\frac{5}{3}} + u^{-\frac{1}{3}}) du$$

$$-\frac{1}{\pi} \left[\frac{3}{8} u^{\frac{8}{3}} + \frac{3}{2} u^{\frac{2}{3}} \right] + C$$

$$-\frac{1}{\pi} \left[\frac{3}{8} (\cot(\pi x))^{\frac{8}{3}} + \frac{3}{2} (\cot(\pi x))^{\frac{2}{3}} \right] + C$$

$$\int \cot^3 x \csc^4 x \, dx$$

Solution 1

$$\int \cot^3 x \csc^4 x \, dx$$

$$\int \cot^2 x \csc^3 x \csc x \cot x \, dx$$

$$\int (\csc^2 x - 1) \csc^3 x \csc x \cot x \, dx$$

$$\text{Let } u = \csc x$$

$$du = -\csc x \cot x \, dx$$

$$-du = \csc x \cot x \, dx$$

$$-\int (u^2 - 1) u^3 \, du$$

$$-\int (u^5 - u^3) \, du$$

$$-\left[\frac{u^6}{6} - \frac{u^4}{4} \right] + C$$

$$-\left[\frac{(\csc x)^6}{6} - \frac{(\csc x)^4}{4} \right] + C$$

Solution 2

$$\int \cot^3 x \csc^4 x \, dx$$

$$\int \cot^3 x \csc^2 x \csc^2 x \, dx$$

$$\int \cot^3 x (\cot^2 x + 1) \csc^2 x \, dx$$

$$\text{Let } u = \cot x$$

$$du = -\csc^2 x \, dx$$

$$-du = \csc^2 x \, dx$$

$$-\int u^3 (u^2 + 1) \, du$$

$$-\int (u^5 + u^3) \, du$$

$$-\left[\frac{u^6}{6} + \frac{u^4}{4} \right] + C$$

$$-\left[\frac{(\cot x)^6}{6} + \frac{(\cot x)^4}{4} \right] + C$$



Text Book Question



$$\int_0^{\frac{\pi}{4}} \sec^6 \theta \tan^6 \theta \, d\theta$$

زویا

فری
X

$$\int_0^{\frac{\pi}{4}} \sec^4 \theta \tan^6 \theta \sec^2 \theta \, d\theta$$

$$\int_0^{\frac{\pi}{4}} (1 + \tan^2 \theta)^2 \tan^6 \theta \sec^2 \theta \, d\theta$$

Let $u = \tan \theta$
 $du = \sec^2 \theta \, d\theta$

$$\int (1 + u^2)^2 u^6 \, du$$

$$\int (1 + 2u^2 + u^4) u^6 \, du$$

$$\int (u^6 + 2u^8 + u^{10}) \, du$$

$$= \frac{u^7}{7} + 2 \frac{u^9}{9} + \frac{u^{11}}{11} = \left[\frac{(\tan \theta)^7}{7} + \frac{2(\tan \theta)^9}{9} + \frac{(\tan \theta)^{11}}{11} \right]_0^{\frac{\pi}{4}}$$

$$= \frac{1}{7} + \frac{2(1)}{9} + \frac{1}{11} - 0$$

$$\int \text{sech}^6 x \, dx$$

$$\int \text{sech}^4 x \, \text{sech}^2 x \, dx$$

$$\int (\text{sech}^2 x)^2 \, \text{sech}^2 x \, dx$$

$$\int (1 - \tanh^2 x)^2 \, \text{sech}^2 x \, dx$$

Let $u = \tanh x$
 $du = \text{sech}^2 x \, dx$

$$\int (1 - u^2)^2 \, du$$

$$\int (1 - 2u^2 + u^4) \, du$$

$$= u - 2 \frac{u^3}{3} + \frac{u^5}{5} + C$$

$$= \tanh x - \frac{2}{3} (\tanh x)^3 + \frac{1}{5} (\tanh x)^5 + C$$

$$\int \frac{\tan^3 x \sec^4 x + \sin x}{\sec^3 x} dx$$

$$\int \left(\frac{\tan^3 x \sec^4 x}{\sec^3 x} + \frac{\sin x}{\sec^3 x} \right) dx$$

$$\int \tan^3 x \sec x dx + \int \frac{\sin x}{\sec^3 x} dx$$

مورفیات

$$\int \tan^2 x \sec x \tan x dx$$

فراہا زویا

$$\int (\sec^2 x - 1) \sec x \tan x dx$$

Let $u = \sec x$
 $du = \sec x \tan x dx$

$$\int (u^2 - 1) du$$

$$\frac{u^3}{3} - u + C_1$$

$$\frac{(\sec x)^3}{3} - \sec x + C_1$$

$$\int \sin x \cos^3 x dx$$

Let $u = \cos x$
 $du = -\sin x dx$
 $-du = \sin x dx$

$$\int u^3 du$$

$$\frac{u^4}{4} + C_2$$

$$-\frac{(\cos x)^4}{4} + C_2$$

$$\Rightarrow \frac{(\sec x)^3}{3} - \sec x - \frac{(\cos x)^4}{4} + C$$

$$\int \frac{\sqrt{\tanh x} + \sinh x}{\cosh^4 x} dx$$

$$\int \left(\frac{\sqrt{\tanh x}}{\cosh^4 x} + \frac{\sinh x}{\cosh^4 x} \right) dx$$

$$\int \frac{\sqrt{\tanh x}}{\cosh^4 x} dx$$



لازم رفیجات !!

$$+ \int \frac{\sinh x}{\cosh^4 x} dx$$

$$\int \sinh x (\cosh x)^{-4} dx$$

$$u = \cosh x$$

$$du = \sinh x dx$$

$$\int \sqrt{\tanh x} \operatorname{sech}^4 x dx$$

$$\int \sqrt{\tanh x} \operatorname{sech}^2 x \operatorname{sech}^2 x dx$$

$$\int u^{-4} du$$

$$\int (\tanh x)^{\frac{1}{2}} (1 - \tanh^2 x) \operatorname{sech}^2 x dx$$

$$\frac{u^{-3}}{-3} + C_2$$

$$u = \tanh x$$

$$du = \operatorname{sech}^2 x dx$$

$$\int u^{\frac{1}{2}} (1 - u^2) du$$

$$\int (u^{\frac{1}{2}} - u^{\frac{5}{2}}) du$$

$$= \frac{2}{3} u^{\frac{3}{2}} - \frac{2}{7} u^{\frac{7}{2}} + C_1$$

$$\Rightarrow \frac{2}{3} (\tanh x)^{\frac{3}{2}} - \frac{2}{7} (\tanh x)^{\frac{7}{2}} + \frac{(\cosh x)^{-3}}{-3} + C$$

$$\int \frac{\sin^5 x + \tan x \sec^2 x}{\tan^2 x} dx$$

$$\int \left(\frac{\sin^5 x}{\tan^2 x} + \frac{\cancel{\tan x} \sec^2 x}{\cancel{\tan^2 x}} \right) dx$$

$$\int \frac{\sin^5 x}{\tan^2 x} dx + \int \frac{\sec^2 x}{\tan x} dx$$

$$\int \frac{\sin^5 x}{\frac{\sin^2 x}{\cos^2 x}} dx$$

$$\int \sec^2 x (\tan x)^{-1} dx$$

$$\int \sin^3 x \cos^2 x dx$$

$$u = \tan x$$

$$du = \sec^2 x dx$$

$$\int \sin^2 x \cos^2 x \sin x dx$$

$$\int u^{-1} du$$

$$\int (1 - \cos^2 x) \cos^2 x \sin x dx$$

$$\int \frac{1}{u} du$$

$$u = \cos x$$

$$du = -\sin x dx$$

$$-du = \sin x dx$$

$$-\int (1 - u^2) u^2 du$$

$$-\int (u^2 - u^4) du$$

$$-\left(\frac{u^3}{3} - \frac{u^5}{5} \right) + C_1$$

$$= \ln |u| + C_2$$

$$= \ln |\tan x| + C_2$$

$$\Rightarrow -\left(\frac{\cos^3 x}{3} - \frac{\cos^5 x}{5} \right) + \ln |\tan x| + C$$



Text Book Question



$$\int \frac{1 - \tan^2 x}{\sec^2 x} dx$$

$$\int \left[\frac{1}{\sec^2 x} - \frac{\tan^2 x}{\sec^2 x} \right] dx$$

$$\int \frac{1}{\sec^2 x} dx - \int \frac{\tan^2 x}{\sec^2 x} dx$$

$$\int \cos^2 x dx - \int \frac{\cancel{\sin^2 x} \cancel{\cos^2 x}}{\cancel{\cos^2 x}} dx$$

$$\frac{1}{2} \int (1 + \cos 2x) dx$$

$$\frac{1}{2} \left[x + \frac{\sin 2x}{2} \right] + C_1$$

$$\int \sin^2 x dx$$

$$\frac{1}{2} \int (1 - \cos 2x) dx$$

$$\frac{1}{2} \left[x - \frac{\sin 2x}{2} \right] + C_2$$

$$\frac{1}{2} \left[x + \frac{\sin 2x}{2} \right] - \frac{1}{2} \left[x - \frac{\sin 2x}{2} \right] + C$$

$$\int \tan^4 x \, dx$$

tan بروحها
 * ما تقرت *
 لو
 فردي زوجي

$$\int \tan^2 x \tan^2 x \, dx$$

$$\int \tan^2 x (\sec^2 x - 1) \, dx$$

$$\int (\tan^2 x \sec^2 x - \tan^2 x) \, dx$$

$$\int \tan^2 x \sec^2 x \, dx - \int \tan^2 x \, dx$$

$$u = \tan x$$

$$du = \sec^2 x \, dx$$

$$\int u^2 \, du$$

$$= \frac{u^3}{3} + C_1$$

$$= \frac{(\tan x)^3}{3} + C_1$$

Again

* يتم تكرار الخطوات *

$$\int (\sec^2 x - 1) \, dx$$

$$\tan x - x + C_2$$

$$\Rightarrow \frac{(\tan x)^3}{3} - (\tan x - x) + C$$

$$\int \tan^5 x \, dx$$

$$\int \tan^3 x \cdot \tan^2 x \, dx$$

$$\int \tan^3 x (\sec^2 x - 1) \, dx$$

$$\int (\tan^3 x \sec^2 x - \tan^3 x) \, dx$$

$$\int \tan^3 x \sec^2 x \, dx - \int \tan^3 x \, dx$$

$$u = \tan x$$

$$du = \sec^2 x \, dx$$

$$\int u^3 \, du$$

$$\frac{u^4}{4} + C_1$$

$$\left(\frac{\tan x}{4}\right)^4 + C_1$$

$$\int \tan x \tan^2 x \, dx$$

$$\int \tan x (\sec^2 x - 1) \, dx$$

$$\int (\tan x \sec^2 x - \tan x) \, dx$$

$$\int \tan x \sec^2 x \, dx$$

$$u = \tan x$$

$$du = \sec^2 x \, dx$$

$$\int u \, du = \frac{u^2}{2} + C_2$$

$$= \frac{\tan^2 x}{2} + C_2$$

$$\ln |\sec x| + C_3$$

$$\Rightarrow \left(\frac{\tan x}{4}\right)^4 - \left[\frac{\tan^2 x}{2} - \ln |\sec x| \right] + C$$

$$\int \cot^5 x \, dx$$

$$\int \cot^3 x \cot^2 x \, dx$$

$$\int \cot^3 x (\csc^2 x - 1) \, dx$$

$$\int (\cot^3 x \csc^2 x - \cot^3 x) \, dx$$

$$\int \cot^3 x \csc^2 x \, dx - \int \cot^3 x \, dx$$

$$u = \cot x$$

$$du = -\csc^2 x \, dx$$

$$-du = \csc^2 x \, dx$$

$$-\int u^3 \, du$$

$$-\frac{u^4}{4} + C_1$$

$$-\frac{(\cot x)^4}{4} + C_1$$

$$-\frac{(\cot x)^4}{4} - \frac{\cot^2 x}{2} - \ln|\sin x| + C$$

$$\int \cot x \cot^2 x \, dx$$

$$\int \cot x (\csc^2 x - 1) \, dx$$

$$\int (\cot x \csc^2 x - \cot x) \, dx$$

$$\int \cot x \csc^2 x \, dx$$

$$u = \cot x$$

$$du = -\csc^2 x \, dx$$

$$-du = \csc^2 x \, dx$$

$$-\int u \, du = -\frac{u^2}{2} + C_2$$

$$-\frac{\cot^2 x}{2} + C_2$$



Try to solve



$$\int \coth^4 x \, dx$$

$$\int \tanh^4 x \, dx$$

$$\int \sin(4x)\sin(6x) dx$$

* الزوايا ليست متشابهة *

$$\frac{1}{2} \int [\cos(6x-4x) - \cos(6x+4x)] dx$$

$$\frac{1}{2} \int [\cos(2x) - \cos(10x)] dx$$

(1) ✓
(2) ✓

$$\frac{1}{2} \left[\frac{\sin(2x)}{2} - \frac{\sin(10x)}{10} \right] + C$$

$$\int_0^{\frac{\pi}{4}} \cos(3x)\cos(5x) dx$$

* الزوايا ليست متشابهة *

$$\frac{1}{2} \int_0^{\frac{\pi}{4}} [\cos(5x-3x) + \cos(5x+3x)] dx$$

$$\frac{1}{2} \int_0^{\frac{\pi}{4}} [\cos(2x) + \cos(8x)] dx$$

(1) ✓
(2) ✓

$$\frac{1}{2} \left[\frac{\sin(2x)}{2} + \frac{\sin(8x)}{8} \right]_0^{\frac{\pi}{4}}$$

$$\frac{1}{2} \left[\left(\frac{\sin \frac{\pi}{2}}{2} + \frac{\sin 2\pi}{8} \right) - 0 \right]$$

$$\frac{1}{2} \left[\frac{1}{2} + 0 \right] = \frac{1}{4}$$



Tricky Question !!



$$\int_0^{\frac{\pi}{2}} \sin x \cos x \cos(4x) dx$$

* الزوايا
لحسابها
متشابهة

X (1)
(2)

$$\sin(2x) = 2 \sin x \cos x$$

÷ 2

$$\frac{1}{2} \sin(2x) = \sin x \cos x$$

$$\int_0^{\frac{\pi}{2}} \frac{1}{2} \sin(2x) \cos(4x) dx$$

$$\frac{1}{2} \int_0^{\frac{\pi}{2}} \sin(2x) \cos(4x) dx$$

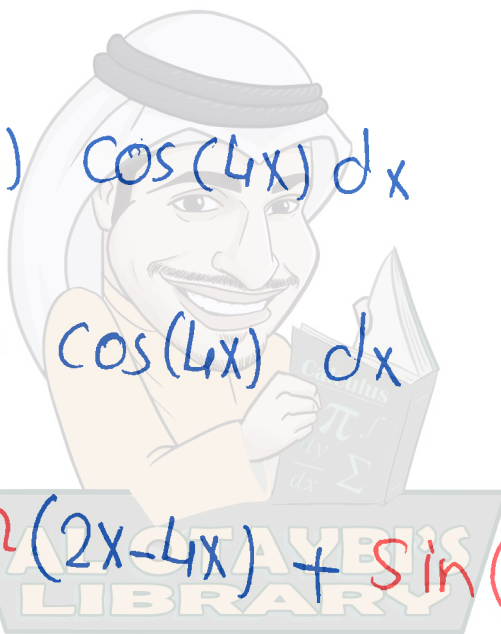
$$\frac{1}{2} \cdot \frac{1}{2} \int_0^{\frac{\pi}{2}} [\sin(2x-4x) + \sin(2x+4x)] dx$$

$$\frac{1}{4} \int_0^{\frac{\pi}{2}} [\sin(-2x) + \sin(6x)] dx$$

$$\frac{1}{4} \left[\frac{\cos(2x)}{2} - \frac{\cos(6x)}{6} \right]_0^{\frac{\pi}{2}}$$

$$\frac{1}{4} \left[\left(\frac{\cos(\pi)}{2} - \frac{\cos(3\pi)}{6} \right) - \left(\frac{1}{2} - \frac{1}{6} \right) \right]$$

$$\frac{1}{4} \left[\frac{-1}{2} - \left(\frac{-1}{6} \right) \right] - \left(\frac{1}{2} - \frac{1}{6} \right)$$





Tricky Question !!



$$\int \cos(3x) \sin^2 x \, dx$$

* الزوايا
للنسبة
متشابهة

$$\int \cos(3x) \frac{1}{2} (1 - \cos 2x) \, dx$$

$$\frac{1}{2} \int (\cos(3x) - \cos(3x) \cos(2x)) \, dx$$

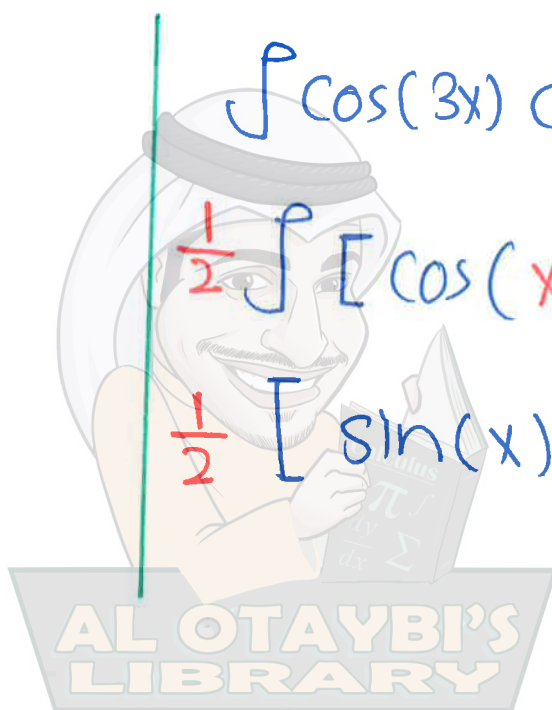
$$\frac{\sin(3x)}{3} + C_1$$

$$\int \cos(3x) \cos(2x) \, dx$$

$$\frac{1}{2} \int [\cos(x) + \cos(5x)] \, dx$$

$$\frac{1}{2} \left[\sin(x) + \frac{\sin(5x)}{5} \right] + C_2$$

$$\Rightarrow \frac{1}{2} \left[\frac{\sin(3x)}{3} - \frac{1}{2} \left[\sin(x) + \frac{\sin(5x)}{5} \right] \right] + C$$



$$\int \cos x \cos(3\sin x) \cos(4\sin x) dx$$

Let $t = \sin x$
 $dt = \cos x dx$

$$\int \cos(3t) \cos(4t) dt$$

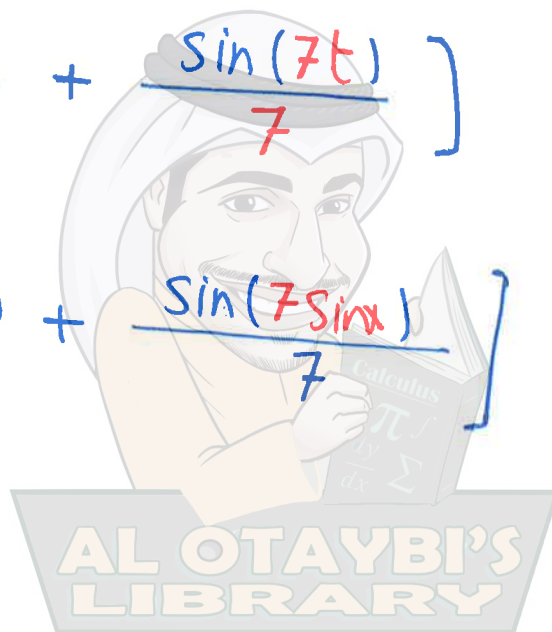
الزوايا ليست
متشابهة

$$\frac{1}{2} \int [\cos(t) + \cos(7t)] dt$$

✓ ①
✓ ②

$$\frac{1}{2} \left[\sin(t) + \frac{\sin(7t)}{7} \right]$$

$$\frac{1}{2} \left[\sin(\sin x) + \frac{\sin(7\sin x)}{7} \right] + C$$





Multiple choice



$\int (\sin(3x))(\cos(5x))dx$ equal to

Oct 2017

(a) $-\frac{\sin(8x)}{16} - \frac{\cos(2x)}{4} + C$

(b) $\frac{\cos(8x)}{16} + \frac{\cos(2x)}{4} + C$

(c) $-\frac{\cos(8x)}{16} + \frac{\sin(2x)}{4} + C$

(d) $-\frac{\cos(8x)}{16} + \frac{\cos(2x)}{4} + C$

(e) $\frac{\sin(8x)}{16} - \frac{\cos(2x)}{2} + C$

$\int \sin(3x) \cos(5x) dx$

$\frac{1}{2} \int [\sin(-2x) + \sin(8x)] dx$

$\frac{1}{2} \int [-\sin(2x) + \sin(8x)] dx$

$\frac{1}{2} \left[\frac{\cos(2x)}{2} - \frac{\cos(8x)}{8} \right] + C$

$\frac{\cos(2x)}{4} - \frac{\cos(8x)}{16} + C$

(d)



المرافق



$$\int \frac{dx}{\cos x - 1}$$



Text Book Question

$$\int \frac{dx}{\cos x - 1} * \frac{\cos x + 1}{\cos x + 1}$$

$$\int \frac{\cos x + 1}{\cos^2 x - 1} dx$$

$$\begin{aligned} \cos^2 x + \sin^2 x &= 1 \\ \cos^2 x - 1 &= -\sin^2 x \end{aligned}$$

$$\int \frac{\cos x + 1}{-\sin^2 x} dx$$

$$= \int \left(\frac{\cos x}{\sin^2 x} + \frac{1}{\sin^2 x} \right) dx$$

$$= \int \left(\cos x (\sin x)^{-2} + \csc^2 x \right) dx$$

$$\int \cos x (\sin x)^{-2} dx = -\cot x$$

Let $u = \sin x$
 $du = \cos x dx$

$$\int u^{-2} du$$

$$\frac{u^{-1}}{-1} = -\frac{1}{u} = -\frac{1}{\cos x}$$

$$\Rightarrow - \left[-\frac{1}{\cos x} - \cot x \right] + C$$



Text Book Question



$$\int \sqrt{1 + \cos 2x} \, dx$$

$$\int \sqrt{1 + \cos 2x} \, dx \quad \frac{\sqrt{1 - \cos 2x}}{\sqrt{1 - \cos 2x}}$$

$$\int \frac{\sqrt{1 - \cos^2 2x}}{\sqrt{1 - \cos 2x}} \, dx$$

$$\int \frac{\sqrt{\sin^2 2x}}{\sqrt{1 - \cos 2x}} \, dx$$

$$\int \sin 2x (1 - \cos 2x)^{-\frac{1}{2}} \, dx$$

Let $u = 1 - \cos 2x$

$$du = 2 \sin 2x \, dx \rightarrow \frac{1}{2} du = \sin 2x \, dx$$

$$\frac{1}{2} \int u^{-\frac{1}{2}} \, du$$

$$= \cancel{\frac{1}{2}} \frac{u^{\frac{1}{2}}}{\cancel{\frac{1}{2}}} + C$$

$$= (1 - \cos 2x)^{\frac{1}{2}} + C$$

هناك حل
آخر

$$\int \sqrt{1 + \cos x} \, dx$$

$$\int \sqrt{1 + \cos x} \cdot \frac{\sqrt{1 - \cos x}}{\sqrt{1 - \cos x}} \, dx$$

$$\int \frac{\sqrt{1 - \cos^2 x}}{\sqrt{1 - \cos x}} \, dx$$

$$\cos^2 x + \sin^2 x = 1$$

$$\sin^2 x = 1 - \cos^2 x$$

$$\int \frac{\sqrt{\sin^2 x}}{\sqrt{1 - \cos x}} \, dx$$

$$\int \sin x (1 - \cos x)^{-\frac{1}{2}} \, dx$$

Let $u = 1 - \cos x$
 $du = \sin x \, dx$

$$\int u^{-\frac{1}{2}} \, du$$

$$= \frac{u^{\frac{1}{2}}}{\frac{1}{2}} + C$$

$$= 2 \sqrt{u} + C$$

$$= 2 \sqrt{1 - \cos x} + C$$



Tricky Question !!



$$\int (1 + \cos 2x)^{\frac{5}{2}} dx$$

$$\int (2\cos^2 x)^{\frac{5}{2}} dx$$

$$\int 2^{\frac{5}{2}} (\cos^2 x)^{\frac{5}{2}} dx$$

$$2^{\frac{5}{2}} \int \cos^5 x dx$$

$$2^{\frac{5}{2}} \int \cos^4 x \cos x dx$$

$$2^{\frac{5}{2}} \int (\cos^2 x)^2 \cos x dx$$

$$2^{\frac{5}{2}} \int (1 - \sin^2 x)^2 \cos x dx$$

Let $u = \sin x$
 $du = \cos x dx$

$$2^{\frac{5}{2}} \int (1 - u^2)^2 du$$

$$\begin{aligned} & \rightarrow 2^{\frac{5}{2}} \int (1 - 2u^2 + u^4) du \\ & 2^{\frac{5}{2}} \left[u - 2\frac{u^3}{3} + \frac{u^5}{5} \right] + C \\ & 2^{\frac{5}{2}} \left[\sin x - \frac{2}{3}(\sin x)^3 + \frac{(\sin x)^5}{5} \right] + C \end{aligned}$$



$$\int (1 - \cos x)^{\frac{3}{2}} dx$$



$$\int (2\sin^2(\frac{x}{2}))^{\frac{3}{2}} dx$$

$$\int 2^{\frac{3}{2}} (\sin^2(\frac{x}{2}))^{\frac{3}{2}} dx$$

$$\begin{aligned} \sin^2 x &= \frac{1}{2}(1 - \cos 2x) \\ \sin^2(\frac{x}{2}) &= \frac{1}{2}(1 - \cos x) \\ 2\sin^2(\frac{x}{2}) &= 1 - \cos x \end{aligned}$$



$$\int \csc x \, dx = \ln |\csc x - \cot x| + C$$

$$\int e^x \tan(e^x) \, dx$$

$$\text{Let } t = e^x \\ dt = e^x dx$$

$$\begin{aligned} \int \tan t \, dt \\ = \ln |\sec t| + C \\ = \ln |\sec(e^x)| + C \end{aligned}$$

$$\int \frac{\csc x \cot x}{5 \csc x + 7} dx$$

$$\left(-\frac{1}{5}\right) \int \frac{-5 \csc x \cot x}{5 \csc x + 7} dx$$

$$\left(-\frac{1}{5}\right) \ln |5 \csc x + 7| + C$$

$$\int x \cos x \, dx$$

$$u = x$$

$$du = dx$$

$$dv = \cos x \, dx$$

$$v = \sin x$$

$$x \sin x - \int \sin x \, dx$$

$$x \sin x + \cos x + C$$

